

ICE CREAM IN A BAG LAB — M5

Chemistry Worksheet • Partner Lab • 20 points

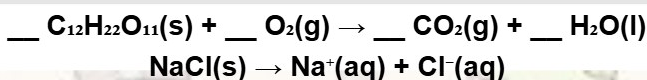
Name(s): _____

Section: _____

Date: _____

Partner(s): _____

LAB CHEMISTRY MODELS / EQUATIONS



The sucrose combustion equation is quantitative practice using the sugar you measured; combustion does NOT happen while making the ice cream.

MEASUREMENTS FOR ONE PAIR

Ingredient / material	Amount
Milk	1/2 cup (120 mL)
Heavy cream	1/2 cup (120 mL)
Sugar (sucrose)	2 tbsp (about 25.0 g)
Vanilla extract	1/2 tsp (2.5 mL)
Ice	about 3 cups (about 700 mL)
Salt (NaCl)	1/3 cup (use 90.0 g for calculations)

PROCEDURE

- Measure 120 mL milk + 120 mL heavy cream + 25.0 g sugar + 2.5 mL vanilla into the SMALL bag. Seal tightly.
- Place about 700 mL ice + 90.0 g salt into the LARGE bag.
- Put the sealed small bag inside the large bag. Shake/turn continuously until the mixture thickens.
- Record observations. Keep salt water outside the food bag; wipe the small bag before opening.

OBSERVATIONS

Stage	Texture / appearance	Coldness / temperature	Approx. time
Before shaking	The mixture in the small bag is a liquid with a creamy white color. The milk and heavy cream are well combined with the sugar and vanilla extract, forming a smooth, uniform liquid mixture with no visible separation or clumps.	The small bag feels cool to room temperature. The ingredients were stored in the refrigerator. The small bag feels very cold to the touch from the surrounding ice and salt mixture. The temperature has dropped significantly below yet, so it remains in liquid form at about -5°C.	The small bag before any shaking begins. About 5 minutes of continuous shaking.
Halfway	The mixture has started to thicken and become slushy. Small ice crystals are forming, giving it a partially frozen texture. It is no longer completely liquid, but it is not yet solid ice cream.	The mixture begins to freeze.	This is approximately halfway through the process of making ice cream in the bag using the ice and salt method.
Finished	Thick, creamy ice cream with a soft-serve consistency. Smooth and scoopable.	Very cold, below freezing temperature. The ice cream is frozen solid enough to hold its shape but still soft enough to eat, approximately -5°C to -8°C.	

QUANTITATIVE + MOLECULAR ANALYSIS

1. Balance the sucrose combustion equation above using the smallest whole-number coefficients. [2 pt]

$\text{C}_{12}\text{H}_{22}\text{O}_{11}(\text{s}) + 12 \text{O}_2(\text{g}) \rightarrow 12 \text{CO}_2(\text{g}) + 11 \text{H}_2\text{O}(\text{l})$

2. Your group used 25.0 g sucrose. Calculate moles of sucrose. Use $M = 342.30 \text{ g/mol}$ and show units. [3 pt]

$n = \frac{m}{M} = \text{molar mass}$

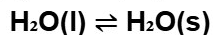
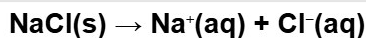
$n = 25.0 \text{ g} \div 342.30 \text{ g/mol}$

3. Using your balanced equation and Question 2, calculate the theoretical moles of CO_2 that would form if that measured sugar were completely combusted. [3 pt]

From the balanced equation, 1 mole of sucrose produces 12 moles of CO_2 .

Mole ratio: $n = 12 \text{ mol CO}_2 / 1 \text{ mol C}_{12}\text{H}_{22}\text{O}_{11}$

LAB CHEMISTRY MODELS / EQUATIONS



Use the outside ice-salt bath and the actual lab measurements below.

4. Assume the 1/3 cup salt used in the large bag had a mass of 90.0 g. Calculate moles of NaCl using 58.44 g/mol. [2 pt]

5. Each mole of NaCl produces approximately 1 mol Na⁺ and 1 mol Cl⁻. Using Question 4, calculate the total moles of dissolved ions produced. [2 pt]

6. Explain why dissolved salt lowers the freezing point of the ice-water mixture and allows the milk/cream mixture to cool enough to freeze. [2 pt]

7. Milk/cream and dissolved sugar contain polar molecules. Identify one important intermolecular force in the mixture and explain what cooling does to molecular motion and texture. [2 pt]

8. Vanilla extract was only 2.5 mL. Is vanilla acting mainly as a reactant, a solvent/solute component, or a flavor mixture in this lab? Explain why there is no new 'vanilla product' in the equation. [1 pt]

9. If a group doubled the sugar but kept the milk, cream, vanilla, ice, and salt the same, predict one likely effect on sweetness and one likely effect on freezing/texture. Explain. [2 pt]

10. Choose ONE controlled variable (milk/cream amount, sugar, vanilla, ice, salt, or shaking time) and explain why keeping it constant makes the experiment more reliable. [1 pt]

Participation: 20 points | Completed worksheet: 20 points | Total lab score: 40 points

4. moles = mass ÷ molar mass

moles = 90.0 g ÷ 58.44 g/mol

moles = 1.54 mol NaCl

5. From Question 4: 1.54 mol NaCl

1.54 mol Na⁺ + 1.54 mol Cl⁻

Total moles of dissolved ions = 3.08 mol

6. Dissolved salt lowers the freezing point of water through freezing point depression. The dissolved ions disrupt ice crystal formation, requiring a lower temperature for the water to freeze. This allows the ice-salt mixture to reach temperatures below 0°C, cold enough to freeze the milk and cream mixture inside the bag.

7. An important intermolecular force is hydrogen bonding between the polar water, sugar, and milk molecules. Cooling decreases molecular motion and kinetic energy, causing molecules to move more slowly. This allows the mixture to solidify and develop a thick, creamy texture as molecules arrange into a semi-solid structure.

8. Vanilla is acting mainly as a flavor mixture component. The small 2.5 mL amount shows it is used for flavoring, not as a reactant. There is no new vanilla product in the equation because vanilla does not chemically react with other ingredients; it simply dissolves into the mixture to add flavor without changing composition.

9. Doubling the sugar would make the ice cream noticeably sweeter because there is more sugar dissolved per volume of mixture. For freezing and texture, the extra sugar would lower the freezing point further, making the mixture harder to freeze completely and resulting in a softer, smoother texture since more sugar interferes with ice crystal growth.

10. Keeping the salt amount constant makes the experiment more reliable because salt controls the freezing point of the ice bath. If different groups used different amounts of salt, their ice baths would reach different temperatures, affecting how well the ice cream freezes. Constant salt ensures all groups have the same cooling conditions for fair comparison.